WIPER MOTOR INSPECTION — GEAR COVER REPLACEMENT

AFFECTED VEHICLES: CERTAIN 1999, 2000 AND 2001 MODEL YEAR WINDSTAR

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OVERVIEW

This procedure gives the detail for installation of the wiper motor gear cover. The wiper module must be removed to install the wiper motor gear cover. You will be instructed to follow the procedures in the workshop manual for wiper module removal and installation.

I. PRE-REPAIR INSPECTION

1. Before performing any repairs, verify that the wiper system is operational on low and high-speed settings, then turn off the wiper system and ensure that the wiper blades travel to the park position.
2. If the motor does not function, perform standard pinpoint diagnostic procedures and repair the system accordingly.

   CAUTION: Wiper system must be parked prior to completing the gear cover replacement procedure to ensure that the internal gear will be in the correct position.

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II. WIPER MOTOR GEAR COVER REPLACEMENT

REMOVE WIPER MOTOR GEAR COVER

1. Remove the wiper module (motor and linkage assembly) from the vehicle and place it on a bench with the back of the wiper motor facing upward. Refer to the service procedures in Section 501-16 of the appropriate Workshop Manual.
2. Using a screwdriver, bend the five gear cover plastic lock tabs away from the motor housing. See Figure 1.

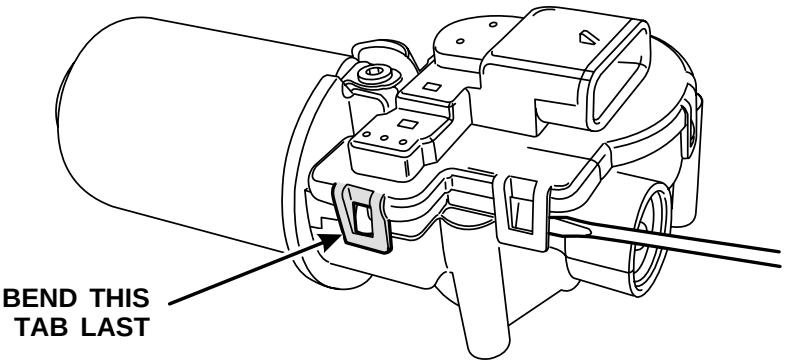


FIGURE 1

3. **CAUTION: The gear cover must be squarely lifted off of the motor housing. Any rocking or sideways motion may bend the three internal electrical connector clips. Pry up around the perimeter of the gear cover to break the sealant from the cover.**

   Once the sealant has been broken loose, squarely lift the gear cover off of the motor housing. Discard the gear cover.
4. Inspect the inside of the wiper motor housing for corrosion.
   - If there are no signs of corrosion inside the housing, continue with this procedure.
   - If the inside of the housing is corroded, it indicates water has entered the wiper motor. Contact the Special Service Support Center at 1-800-325-5621 for further instruction.

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REMOVE SEALANT

CAUTION: Do not use solvents or compressed air to remove the sealant from the motor gear housing. Any solvent entering the gear cavity may damage the motor. Also, remove all large pieces of sealant that enter the gear housing cavity during cleaning. Small particles of sealant left in the cavity are acceptable.

NOTE: It is not necessary to remove the existing sealant attached to the vertical surfaces inside the gear housing.

1. Remove the existing sealant from the top surface of the gear housing and around each of the five mounting lug areas. To avoid damaging the motor, extra care must be taken when removing sealant from the area around the internal electrical connector clips. See Figure 2.

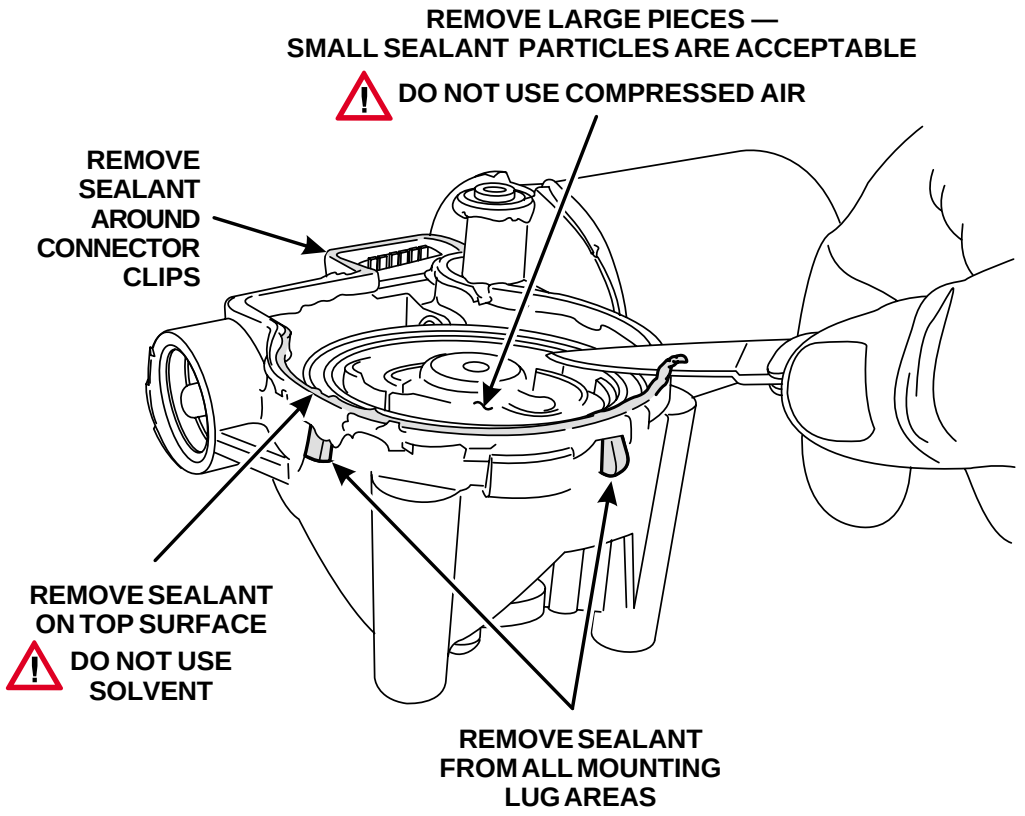


FIGURE 2

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INSPECT MOTOR ELECTRICAL CONNECTOR CLIPS

1. Inspect the three (3) internal electrical connector clips. **All three clips MUST be aligned against the plastic terminal holder.** If any of the clips are misaligned or touching the motor housing, they must be straightened and positioned as shown in the illustration. See Figure 3.

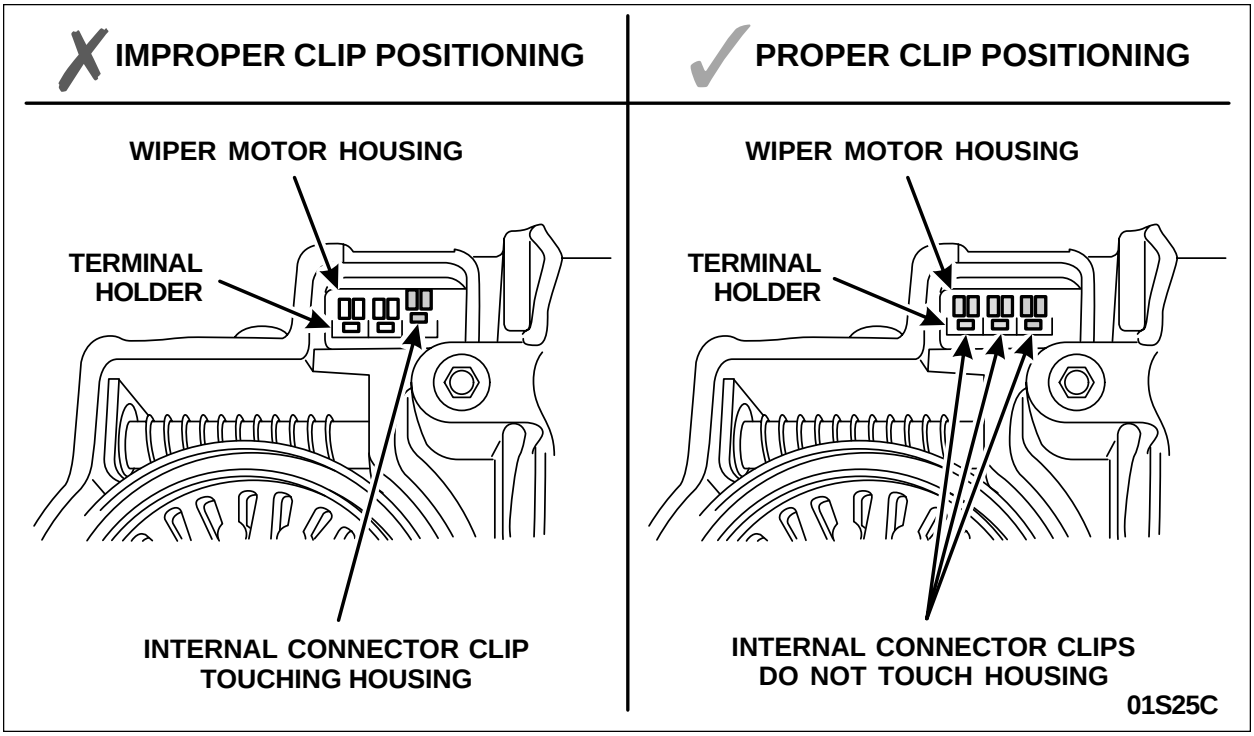


FIGURE 3

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PREPARE REPLACEMENT COVER

1. **IMPORTANT: Using your thumb, bend locking tab "E" approximately 60-90 degrees inward (see Figure 4) then bend it back approximately to the position illustrated in Figure 5. This will improve the locking tab's engagement with the motor gear housing.**

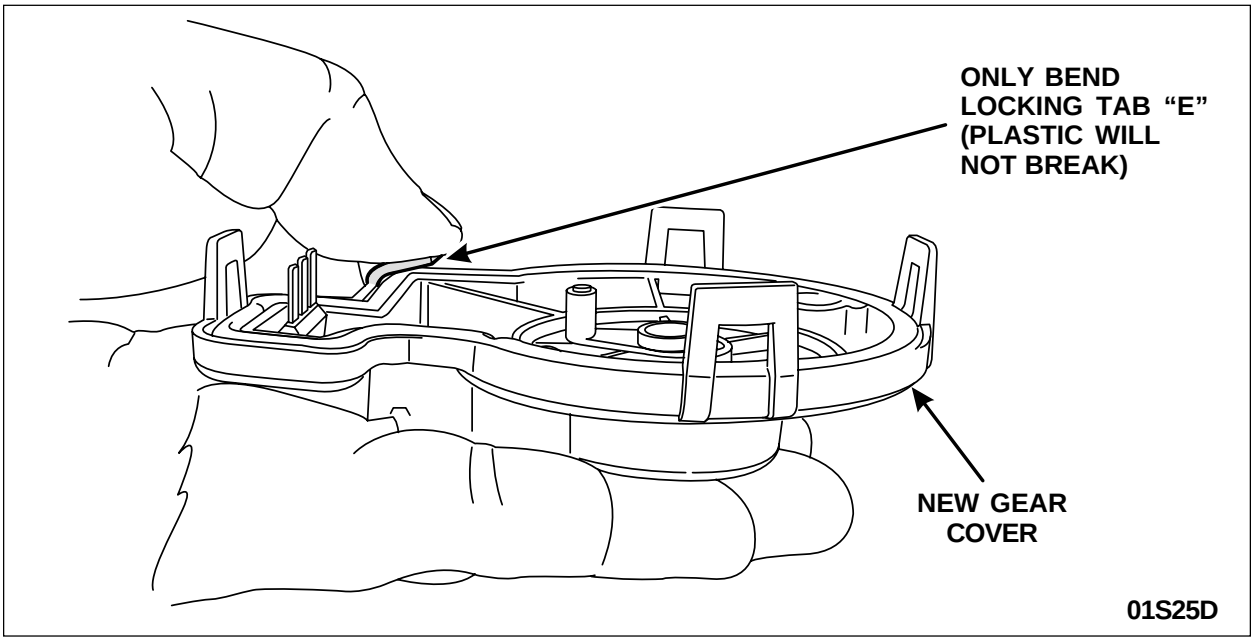


FIGURE 4

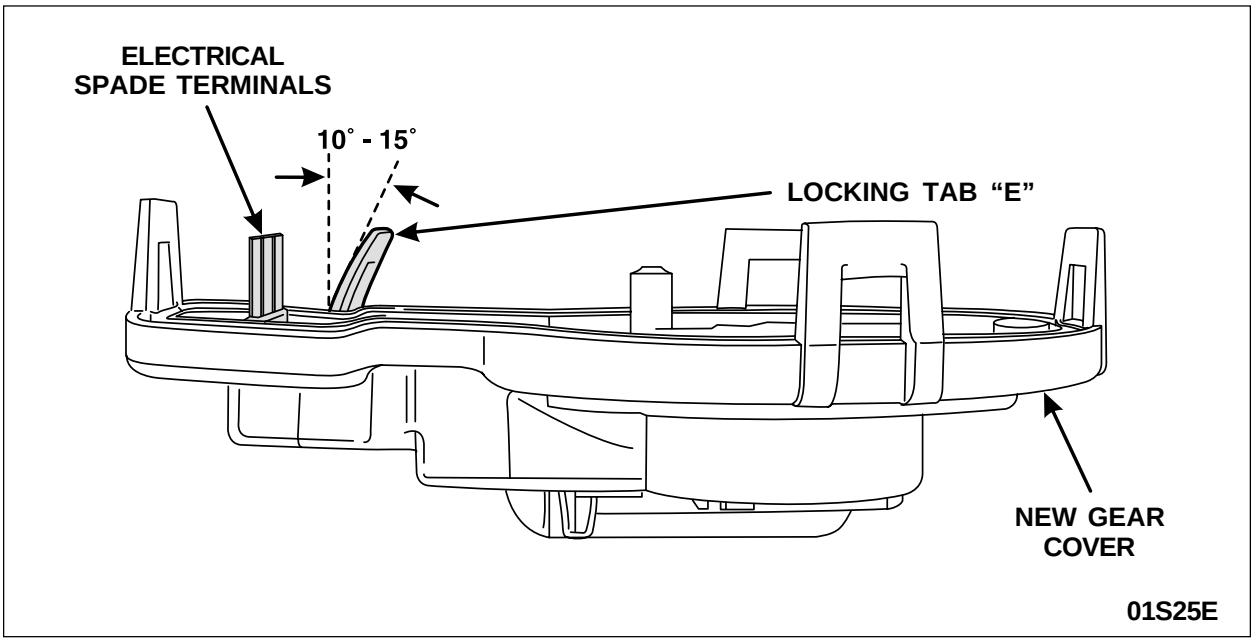


FIGURE 5

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2. Verify that the three (3) electrical spade terminals are not bent. If necessary, straighten them so that they are perpendicular to the cover as illustrated in Figure 6.
3. Make sure that the foam gasket is properly seated in the ***new*** cover as shown in Figure 6.
4. **Make sure that the white cam follower arm is positioned as shown in Figure 6.**

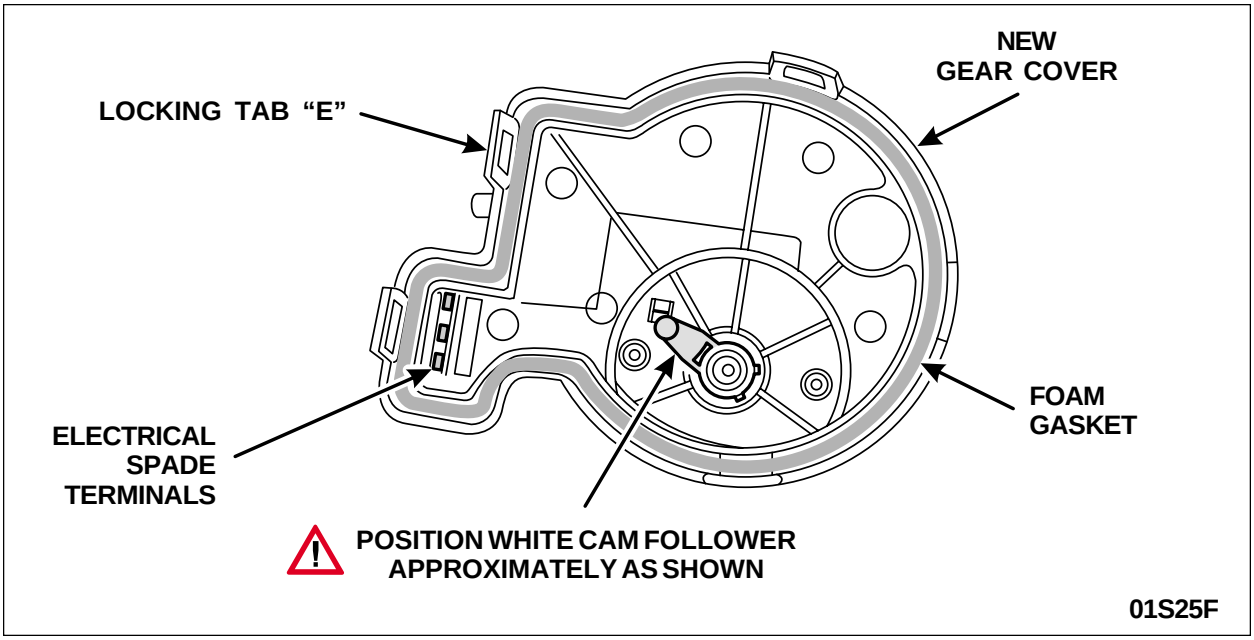


FIGURE 6

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INSTALL REPLACEMENT COVER

1. **IMPORTANT: Position the cover on the gear housing so that each plastic locking tab is outside of the gear housing cavity and is resting over the mounting lugs. Make sure that the electrical spade terminals are lined up with the connector clips. See Figure 7.**
2. Using two hands, push the gear cover downward onto the gear housing uniformly, to ensure that the electrical terminals remain in alignment. Push down firmly until all five locking tabs are latched correctly. See Figure 8.
3. **CAUTION: If any of the locking tabs cannot be correctly latched as shown in Figure 9, replace the complete motor assembly. DO NOT attempt to remove the cover and reinstall it.**

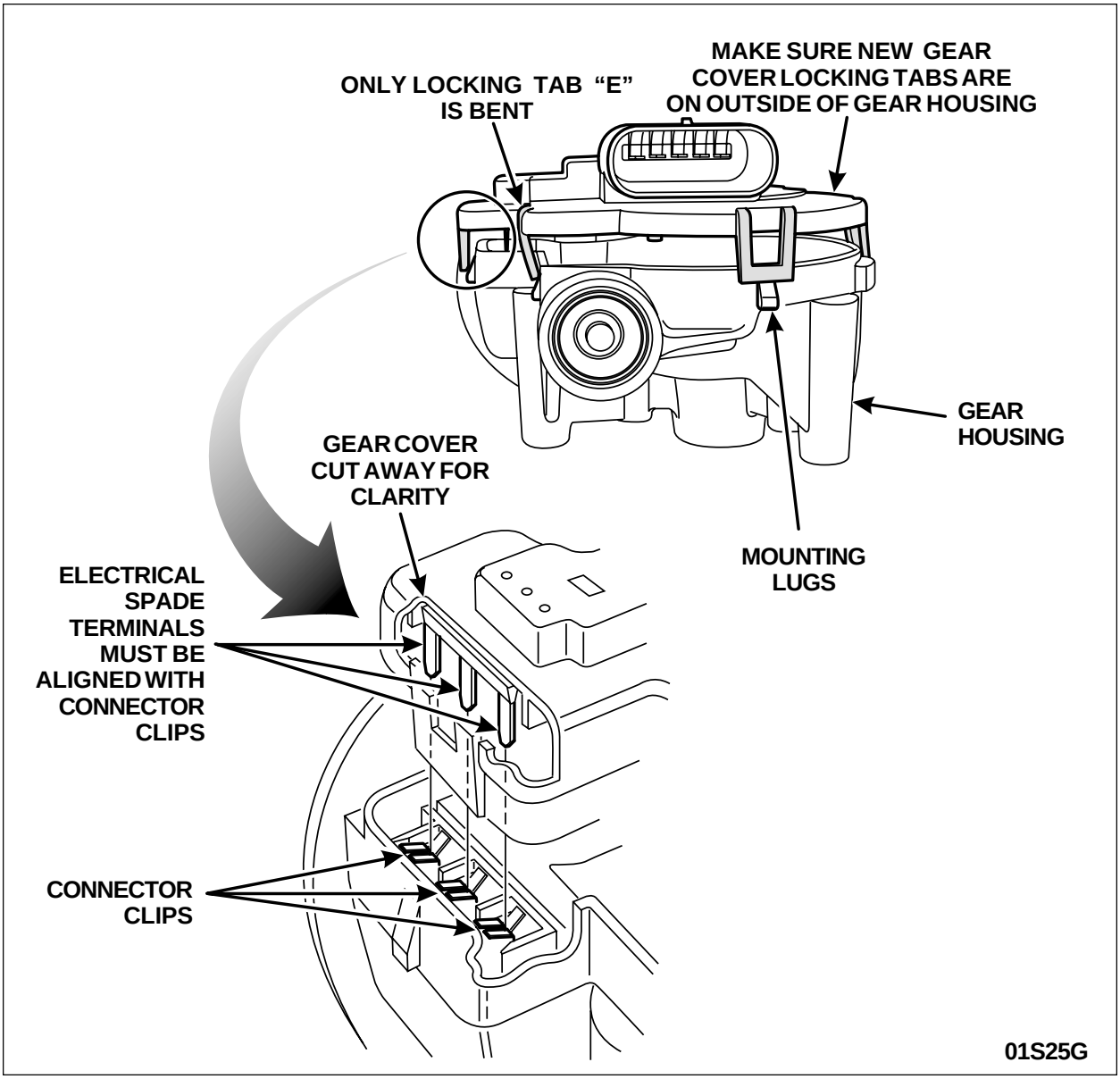


FIGURE 7

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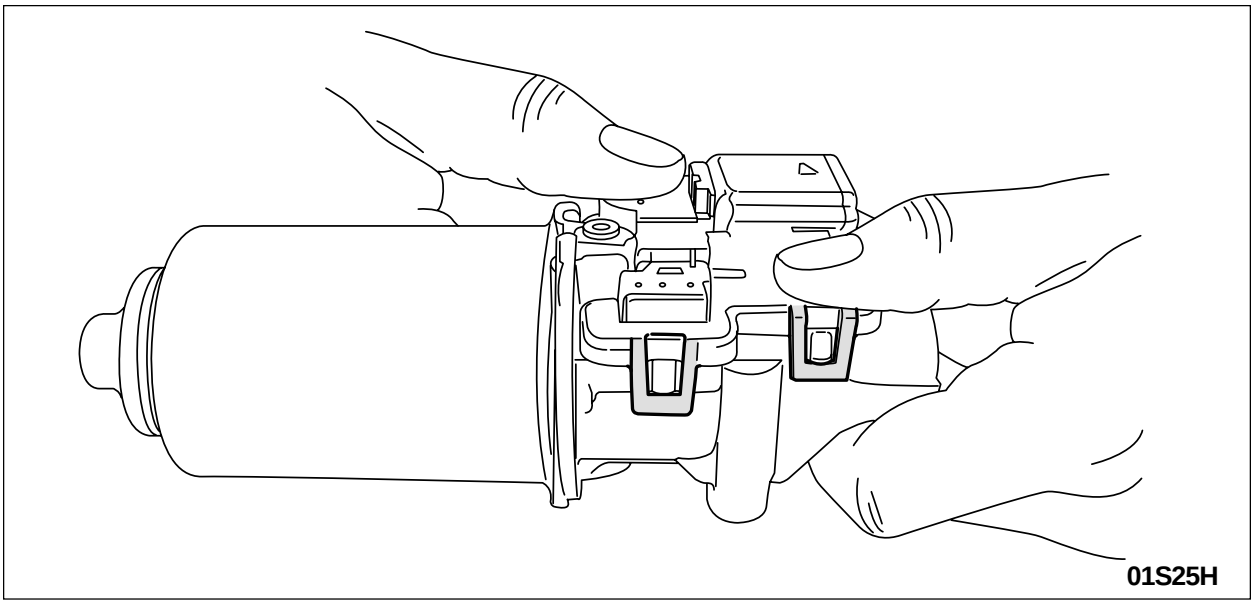


FIGURE 8

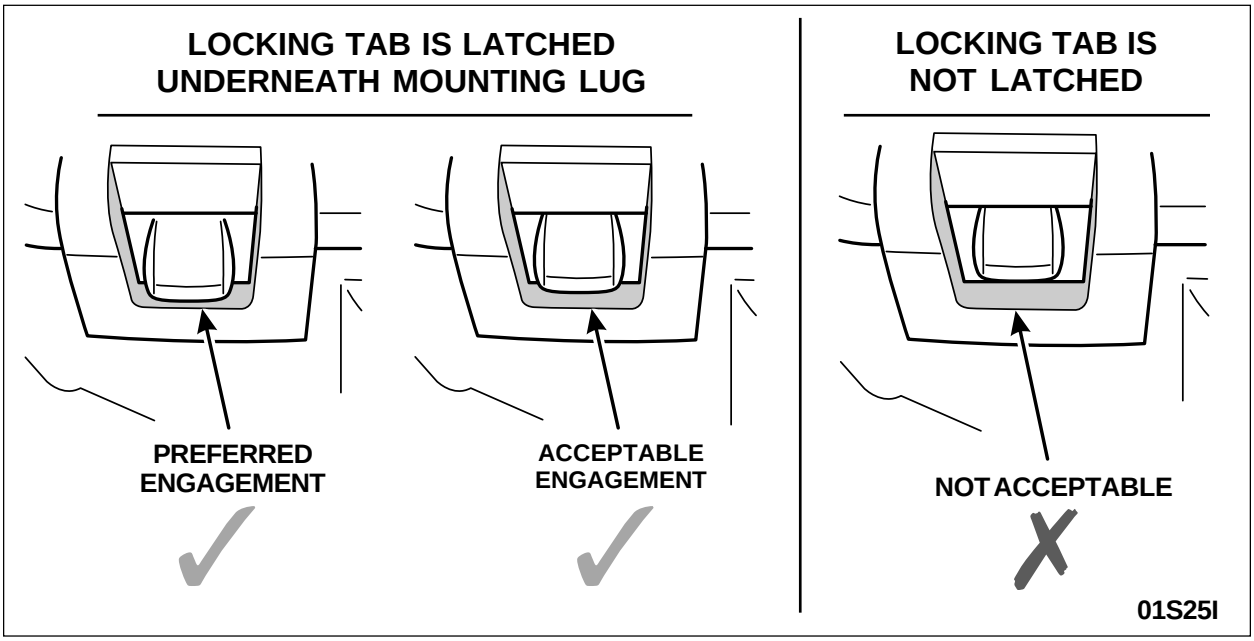


FIGURE 9

III. VERIFY WIPER MOTOR OPERATION

1. Install the wiper module. Refer to the procedures in Sections 501-02 and 501-16 of the appropriate Workshop Manual.
2. Check the operation of the wiper motor (high speed, low speed and park).

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